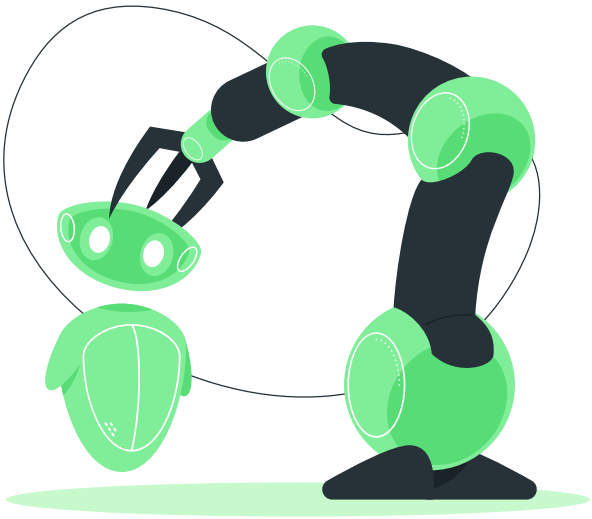
Robotics
CS 3015

Umarah Qaseem

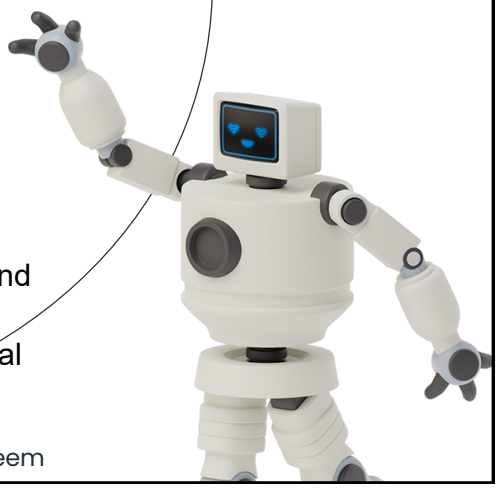


1

- Actuators
- Types of actuators and working
- Artificial Muscles
- Electric Motors
- Gearbox, Sequential Encoders
- H-bridge circuit of motors

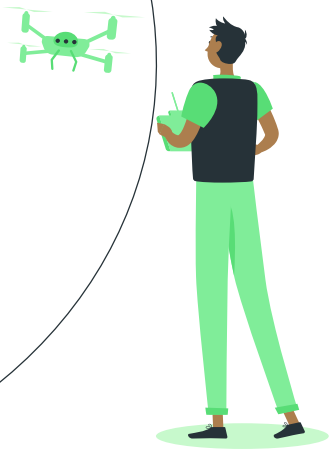
16

- Dynamics and Control
- Control Theory
- Feedback Systems and their types
- Open loop and closed loop systems
- Laplace Transform
- Mathematical modelling of a system and transform equation with Example
- Representation of a system, Differential model and transform model
- Control Theory for robotics



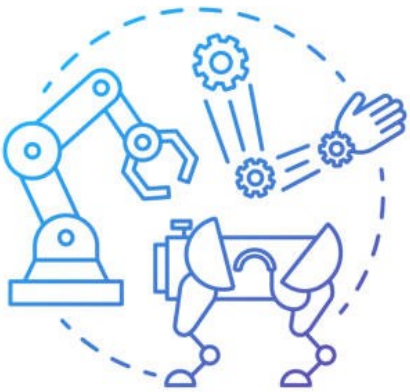
2

Dynamics and Control



Robotics – Umarah Qaseem

4

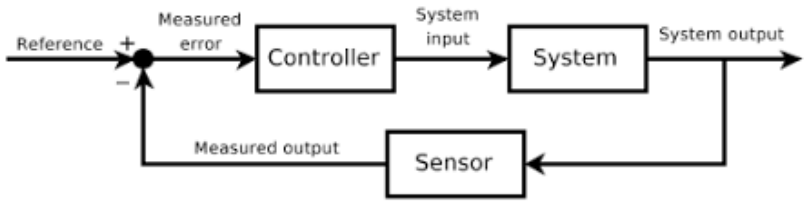


Dynamics and control

5

Review – Control System

Have you studied the subject
"Linear Control Systems" or "Control Theory"?



6

Control Theory for robotics, Feedback Systems



Robotics – Umarah Qaseem

7

Introduction

- Till now
  - We know the desired pose
  - We can calculate the joint required to achieve the desired pose using inverse kinematics

$$\xi_E \longrightarrow q$$

  - We can generate intermediate poses from current to desired pose
- How does a robot move?
- How to control each individual joint?

8

Mechatronics problem

- Trying to control the position of the actuator or a motor
- Very common problem in mechatronics
- Also known as motion control system

9

Motion Control System

- Controlling the position of a motor

- printer: print head, paper feed
- wheels of a robot vacuum cleaner
- moving the heads in a CD/DVD/hard disk
- camera lens control, focus or zoom
- rotor speed of a quadcopter
- etc.

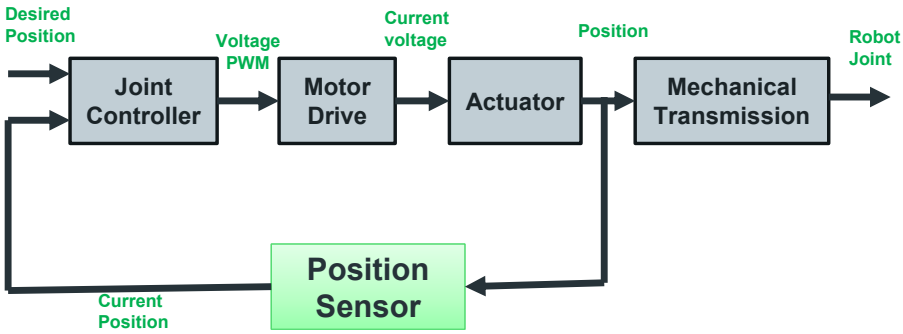
10

What is joint control problem?

- How to make a joint adopt a particular angle?
- Control system will do that for us.
- Joint has a motor (actuator) and some sensors.

11

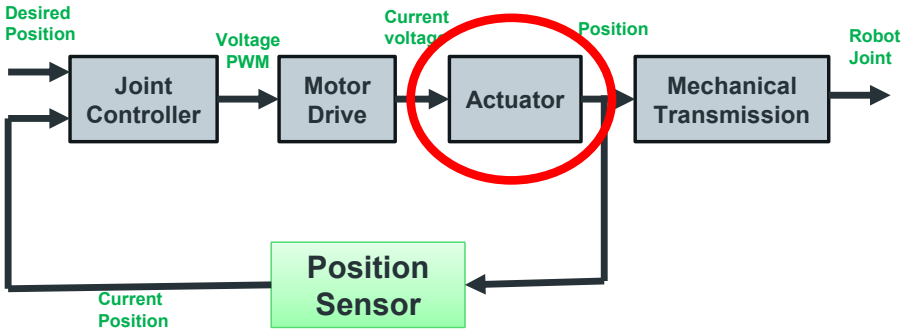
Joint Control Architecture



Discussion

12

Joint Control Architecture



13

Home study

- **Best if you have studied** control systems
- Laplace transform
- Poles of dynamic system

- **Study the following:**
- Motors
- Gearbox
- Embedded controllers
- Encoders

14

Actuators

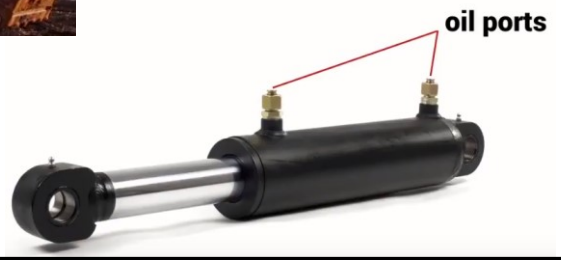
- **Definition**: A device that causes linear or rotary motion
- **Output**: Speed/Torque
- **Types**
  - Hydraulic (compressed oil causes motion)
  - Pneumatic (compressed air causes motion)
  - Electric (electric current and magnets cause the motion)

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Hydraulic Actuators



Converts Oil Pressure to linear force

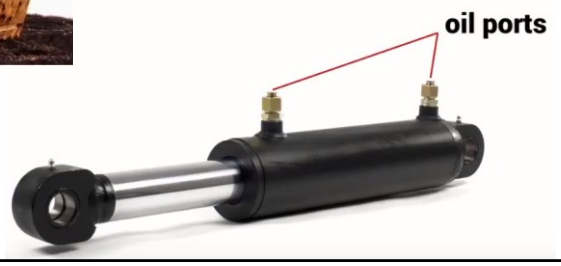


16

Hydraulic Actuators



- Used on heavy machinery
- Generate high forces



17

Work site of the new building under construction in FAST

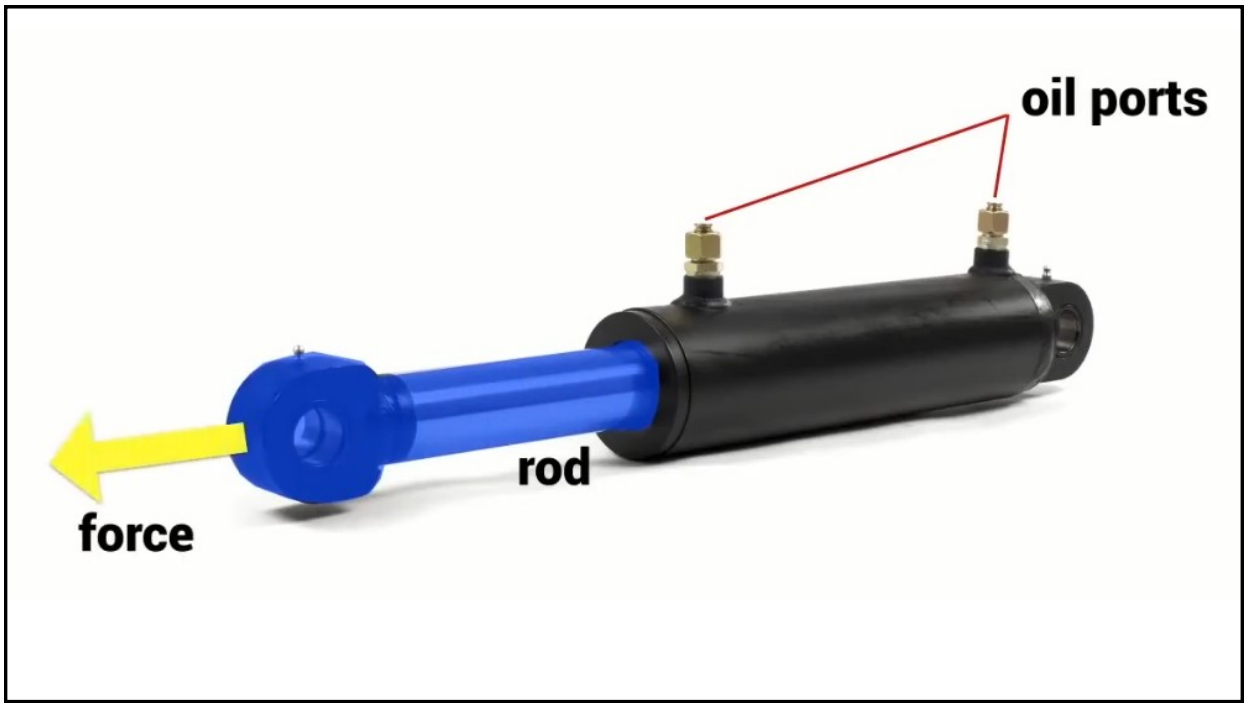


18

Work site of the new building under construction in FAST

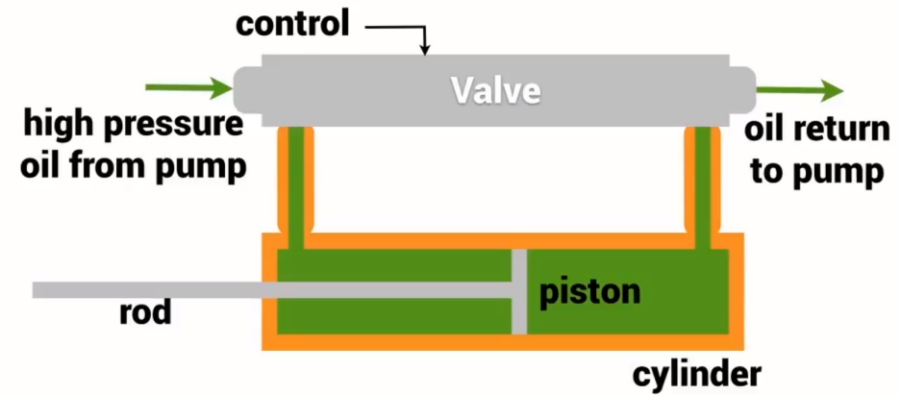


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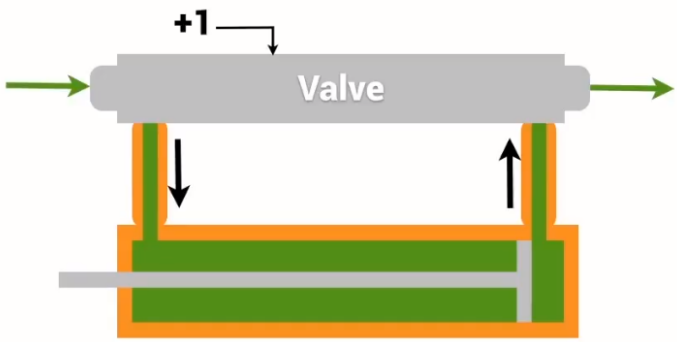
Hydraulic Actuators



21

Hydraulic Actuators

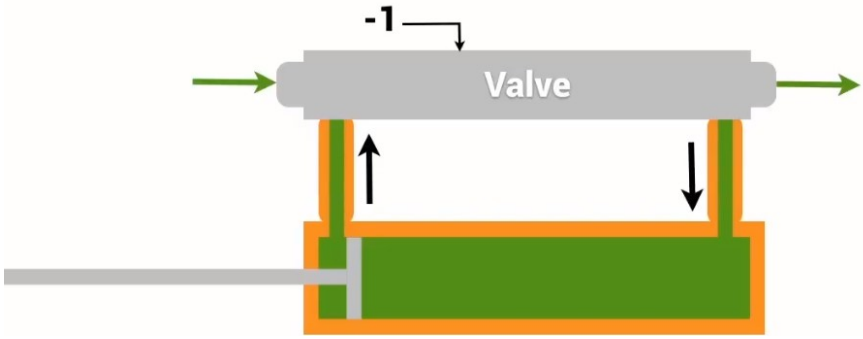
A positive control signal moves the piston towards the right side



22

Hydraulic Actuators

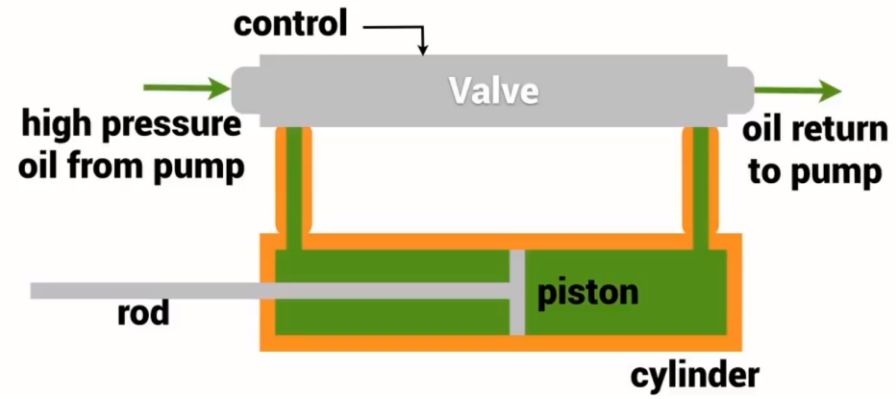
A negative control signal moves the piston towards the left side



23

Hydraulic Actuators

How to lock the piston in one position?



24

Hydraulic Actuators

Linear hydraulic actuators are most common but rotary hydraulic actuators also exist such as oil driven turbine.



25

Pneumatic Actuators

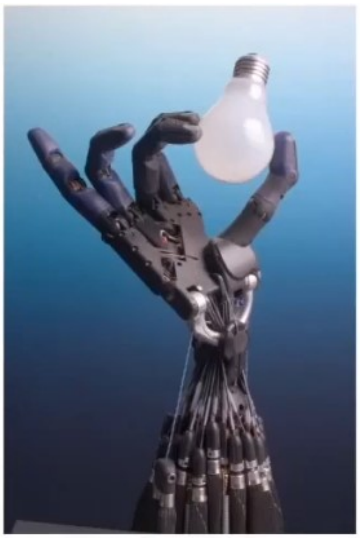
- Converts air pressure to a linear force
- Very similar to hydraulic except that it involves air/gases
- Alternate technology that converts air pressure to linear force is called as artificial muscle (no piston)



27

Artificial Muscle





28

Artificial Muscle



Problem?

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Artificial Muscle



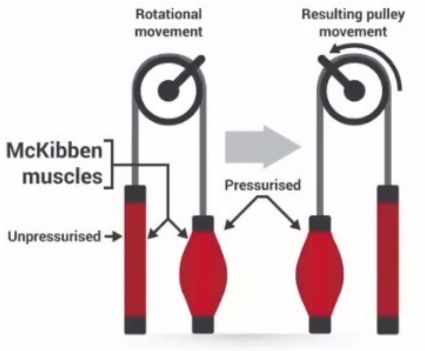
They can pull and can't push.
Like our own muscles

30

Artificial Muscle



Used in antagonistic pairs



31

Electric Motors

- **Principle**: Like poles repel each other, whereas as unlike poles attract each other
- Commutation trick is applied to use the above property to construct a motor

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Electric Motors

- Important components of a Motor
- Stator: stationary electromagnet of the motor
- Rotor: rotating electromagnet of the motor
- Commutator: switched the polarity of the rotor resulting in reversing of the direction of magnetic field

33

Commutation

- What is it?

34

Electric Motors



$$\tau = K_m i$$

35

Electric Motors

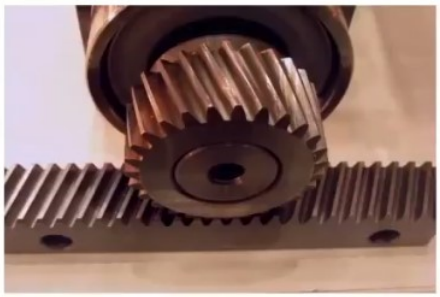
- Compact and reliable
- Converts electric current flowing in the motor to a torque applied to output shaft of the motor and it is a linear relationship

- Alternative view of motor is that it converts an applied voltage to a rotational velocity ω
- How electric motors are used in practice?

$$\tau = K_m i$$

36

Electric Motors

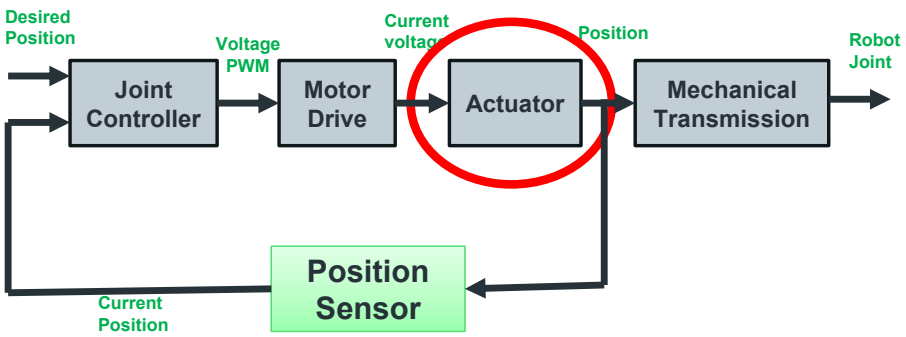




- Convert rotary motion to linear
- Linear actuator

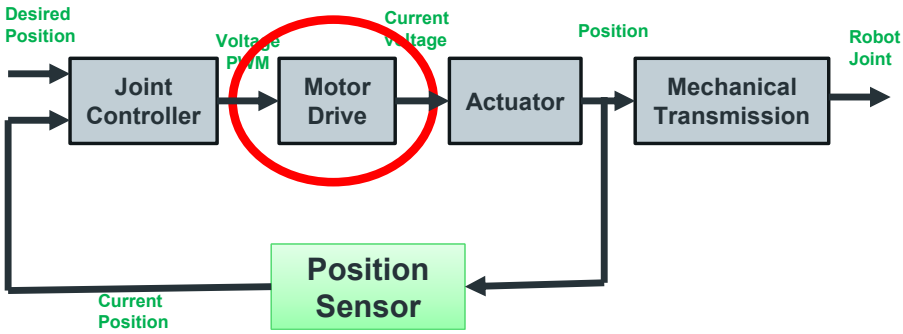
37

Joint Control Architecture



50

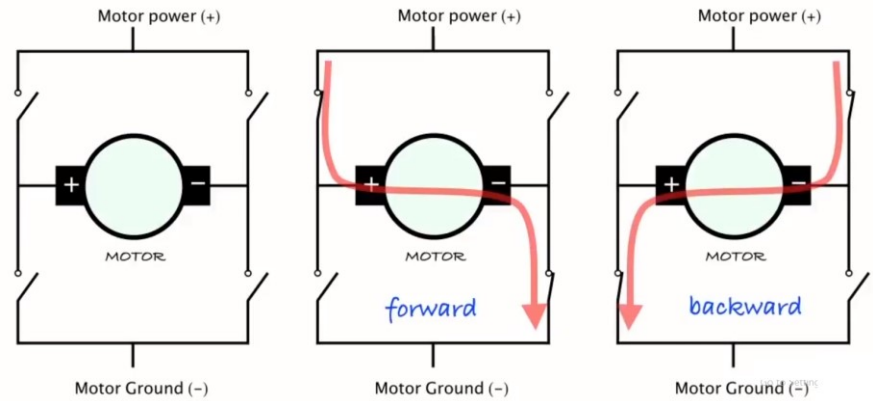
Joint Control Architecture



51

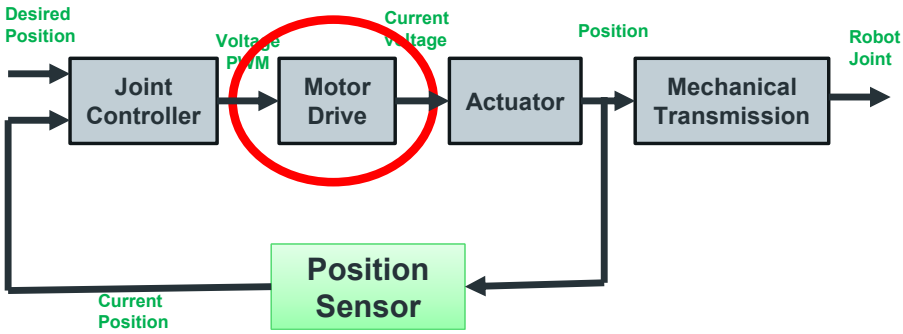
Electric Motor Voltage Control

Motor Drive H-Bridge Circuit



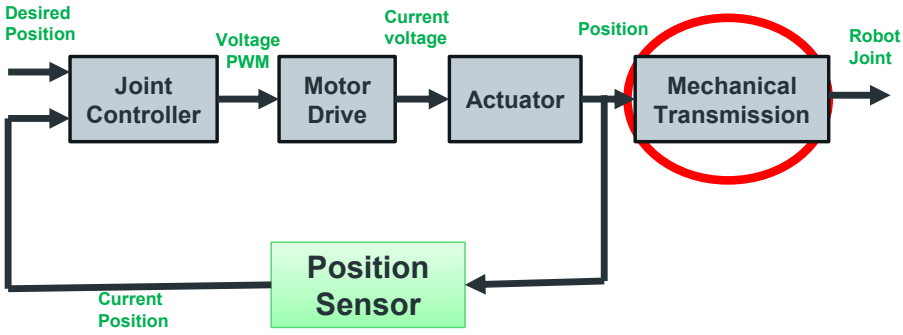
52

Joint Control Architecture



56

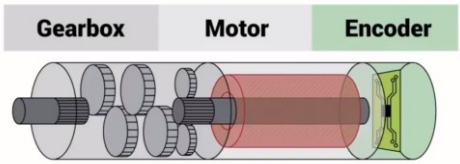
Joint Control Architecture



57

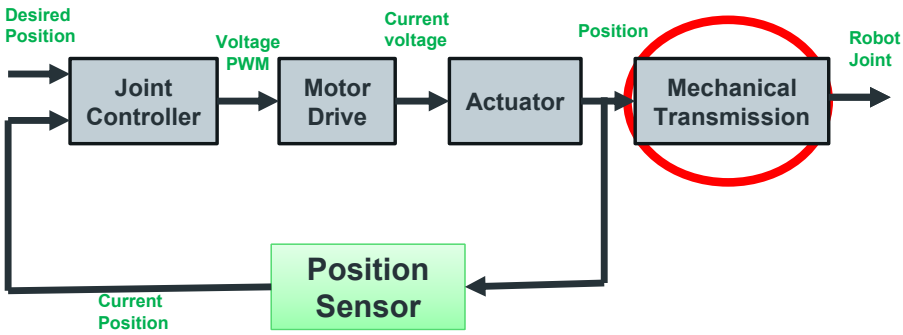
Gearbox

- Electric motors have low torque but high rotational speed
- Gearboxes are introduced to balance speed and torque as per the application requirements
  - Apart from its advantage, there are losses in form of heat and sound and therefore there is always inefficiency factor



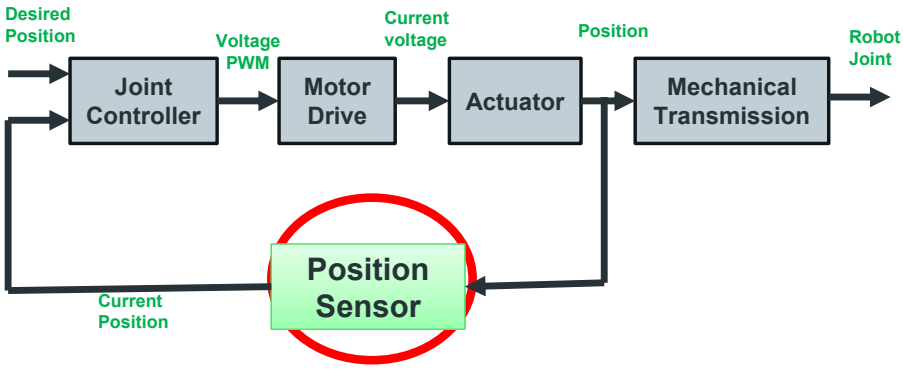
58

Joint Control Architecture



64

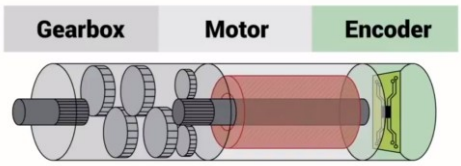
Joint Control Architecture



65

Position Sensor

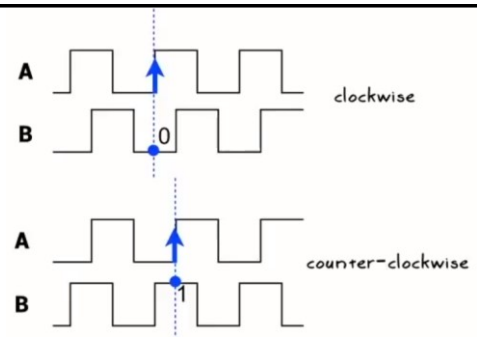
- How to sense the motor angle?
- Incremental Encoders are most commonly used for position sensing in motors

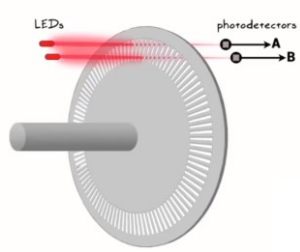


66

Position Sensor

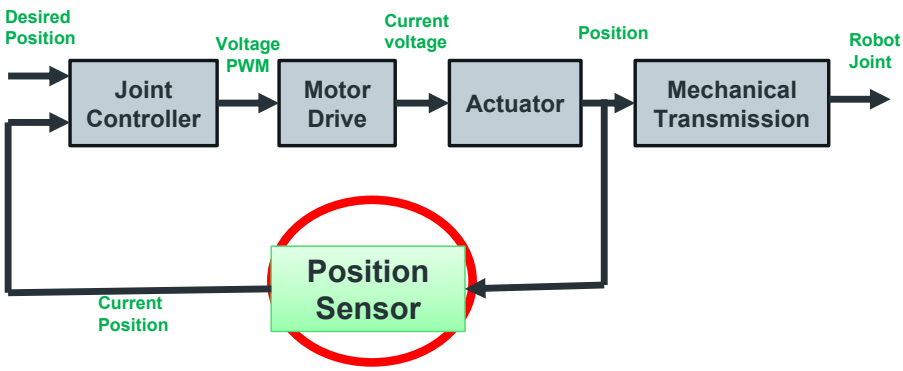
- Incremental Encoder
  - Very simple and cheap
  - Provides direction and rotational speed (pulses per second)
  - Pulse count gives the position
  - Can be easily interfaced with microcontroller





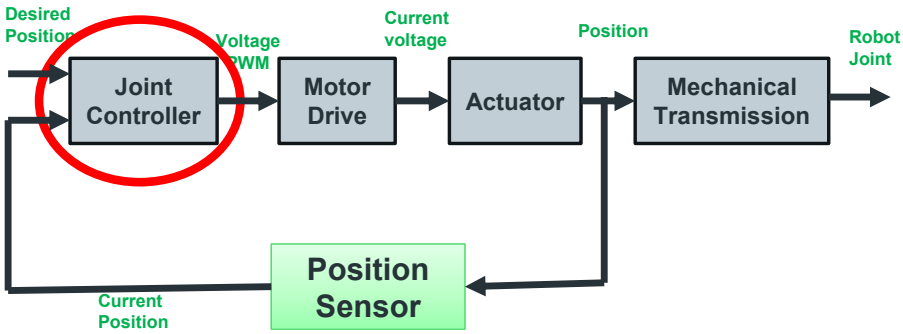
67

Joint Control Architecture

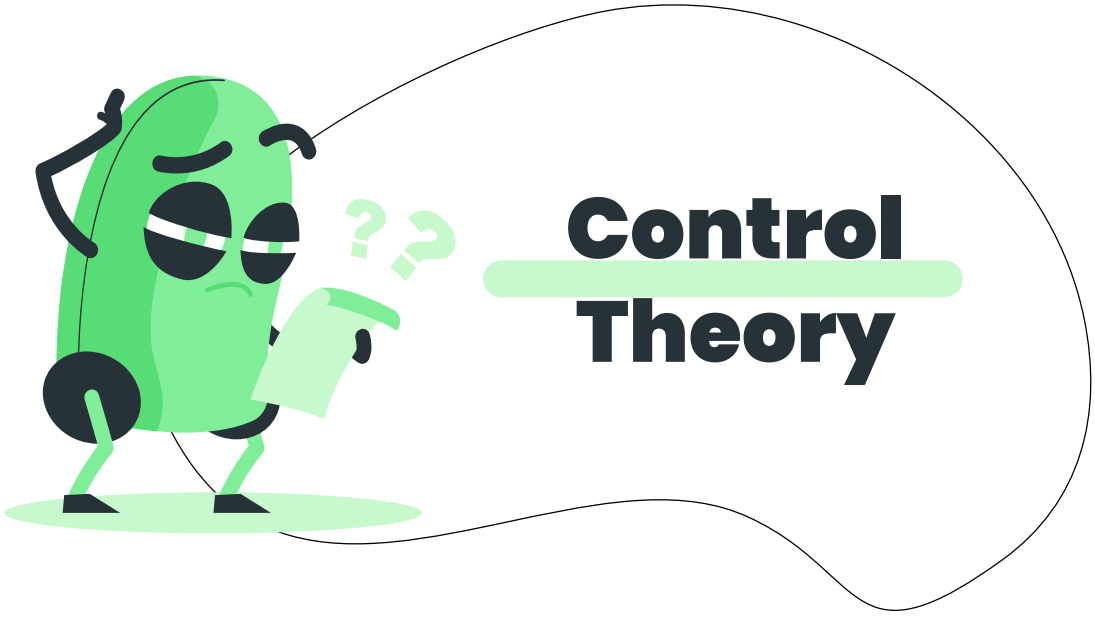


68

Joint Control Architecture

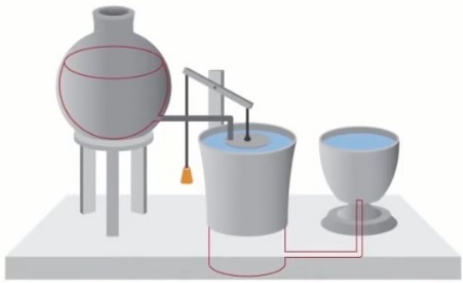


69



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Water Control in History



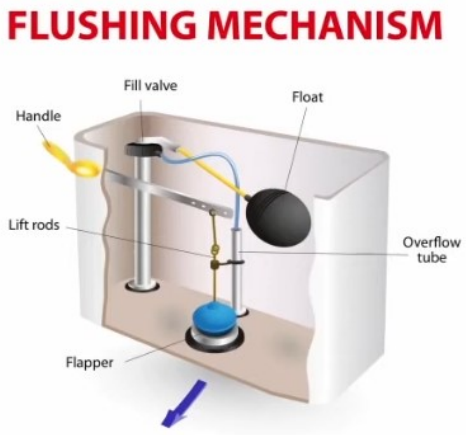
Float regulator of Heron

- Maintains constant **desired** water level in the rightmost vessel
  - ➥ leads to constant flow rate for a water powered clock

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Water Control in history related to a modern mechanism





- Fill the tank until the **desired** water level is achieved
  - ➥ level < desired → water flows
  - ➥ level ≥ desired → no flow

72

Controlling wind power



- The fan tail keeps a windmill pointed into the wind
  - ➥ invented by Edmund Lee in 1754

73

What is a control system?

- A control system consists of sub systems that is used to control the behavior (output) of a device or process

- E.g
- An air conditioner maintaining the temperature of the room
- A car moving from rest to a speed of 120 Km/h
- A lift moving upwards

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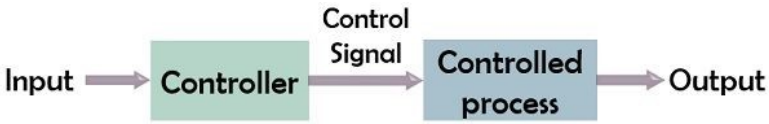
Types

- Two types
- Open loop
- Closed loop

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Open loop control system

- Control action is totally independent of the output of the system



Open Loop Control System

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Open loop control system

- Control action is totally independent of the output of the system
- Accuracy depends on experience of the user
- Example
- Immersion water heating rod.
- Hotplate
- Stove

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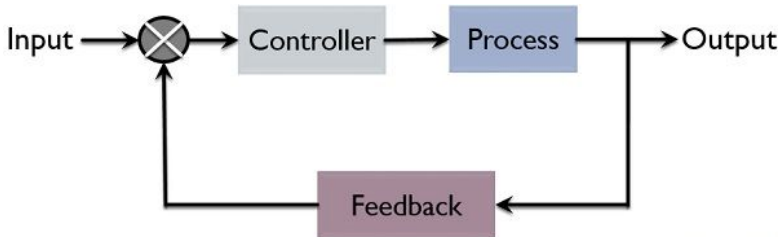
Open loop control system

- Advantages:
- Simple in construction and Design
- Economic
- Convenient to use when the output is difficult to measure
- Disadvantages:
- Poorly equipped to handle disturbances
- Not reliable

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Closed loop control system

- Output is measured continuously and is fed back to the system



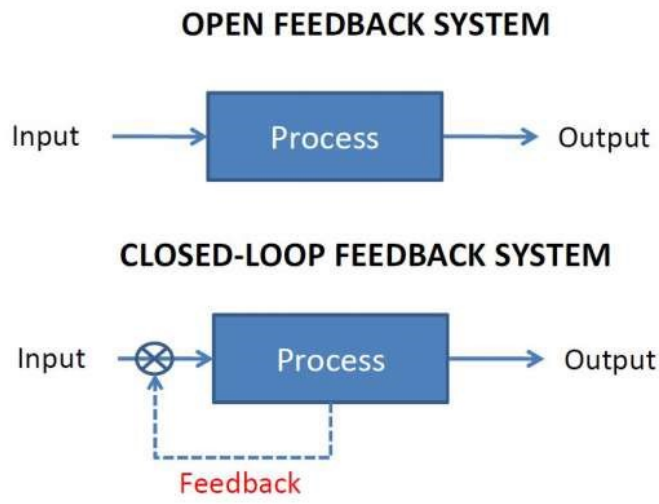
81

Closed loop control system

- Output is measured continuously and is fed back to the system
- Presence of feedback compensates for disturbances and improves the accuracy of the system
- Example
- Air conditioner
- Geyser

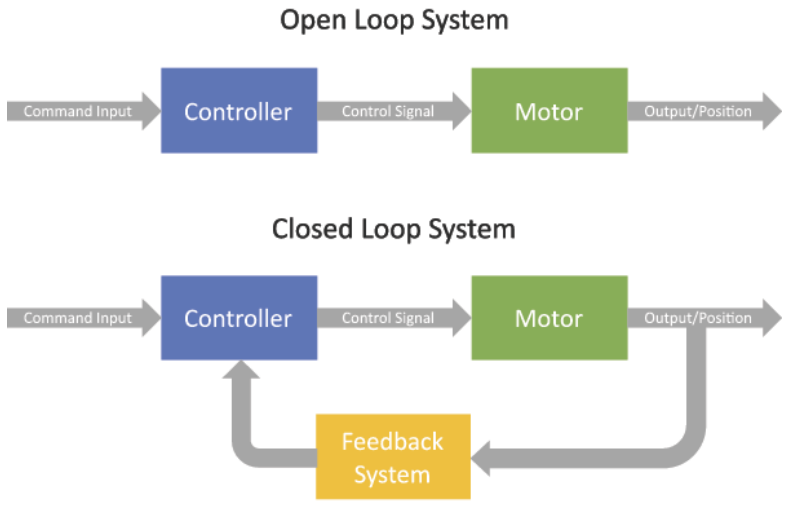
82

Open loop vs Closed loop



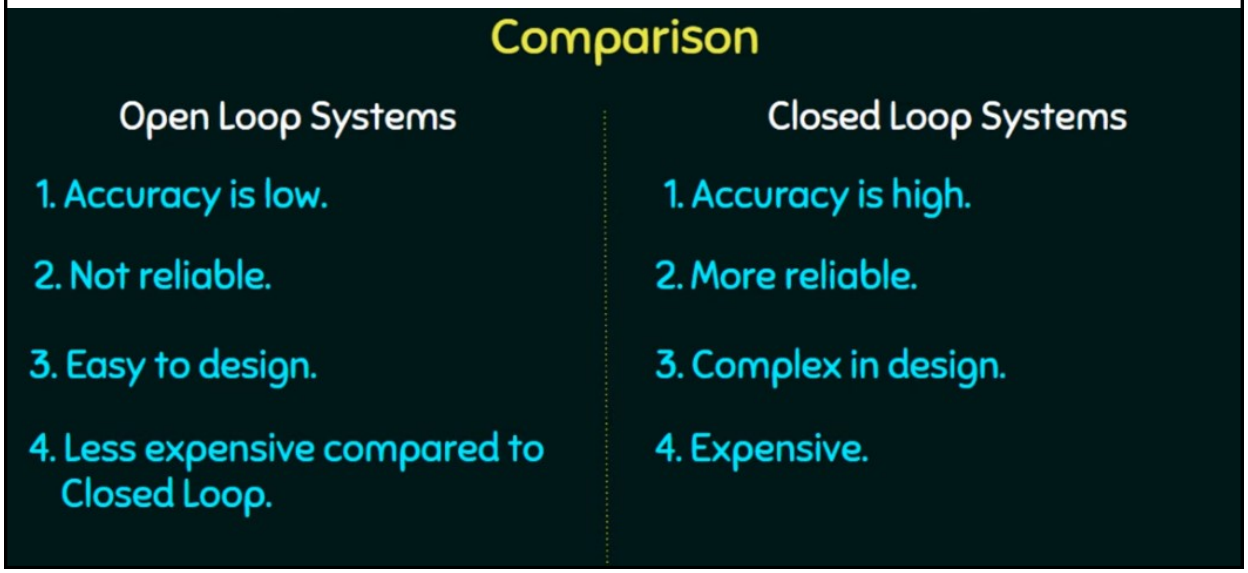
83

Open loop vs Closed loop



84

Open loop vs Closed loop



85

Types of closed loop/feedback CS

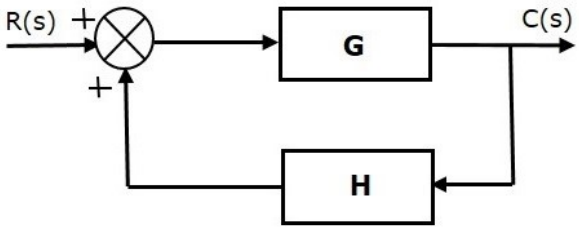
- Two types
- Positive
- Negative

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Positive Feedback

- The positive feedback adds the reference input, R(s) and feedback output.
- the transfer function of positive feedback control system is,

$$T = \frac{G}{1-GH}$$

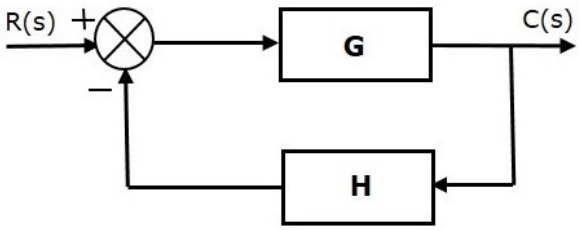


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Negative feedback

- Negative feedback reduces the error between the reference input, R(s) and system output.
- Transfer function of negative feedback control system is

$$T = \frac{G}{1+GH}$$



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Transfer function

- A transfer function is a ratio of a system's output to its input.

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Mathematical model of Control Sys

- The control systems can be represented with a set of mathematical equations known as **mathematical model**.
- These models are useful for analysis and design of control systems.
- Analysis of control system means finding the output when we know the input and mathematical model.
- Design of control system means finding the mathematical model when we know the input and the output.

90

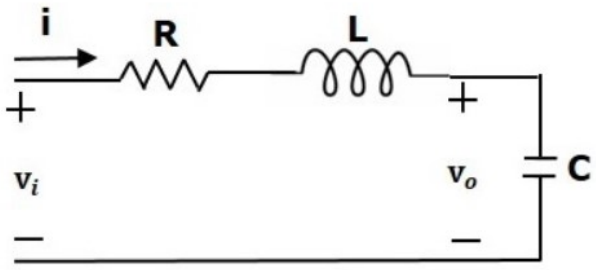
Mathematical model of Control Sys

- We will study two models
- Differential equation model
- Transfer function model

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Differential Equation Model with (Example 1)

- Consider the following electrical system as shown in the following figure. This circuit consists of resistor, inductor and capacitor. All these electrical elements are connected in series. The input voltage applied to this circuit is vi and the voltage across the capacitor is the output voltage vo.



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Differential Equation Model with (Example 1)

- Solved on board

93

Laplace Transform

- Converts a time domain signal to frequency domain

$$\dot{f}(t) = \frac{df(t)}{dt} \longrightarrow sF(s)$$

95

Transfer Function Model

- Laplace Transform of Output / Laplace Transform of Input

$$i.e., Transfer\ Function = \frac{Y(s)}{X(s)}$$

96

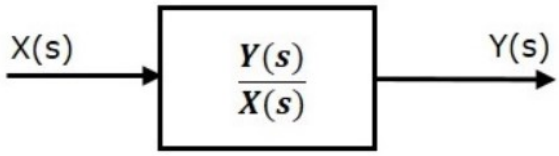
Transfer Function Model

- Transfer function model is an s-domain mathematical model of control systems.
- The Transfer function of a Linear Time Invariant (LTI) system is defined as the ratio of Laplace transform of output and Laplace transform of input by assuming all the initial conditions are zero.
- So If x(t) and y(t) are the input and output of an LTI system, then the corresponding Laplace transforms are X(s) and Y(s).
- Therefore, the transfer function of LTI system is equal to the ratio of Y(s) and X(s)

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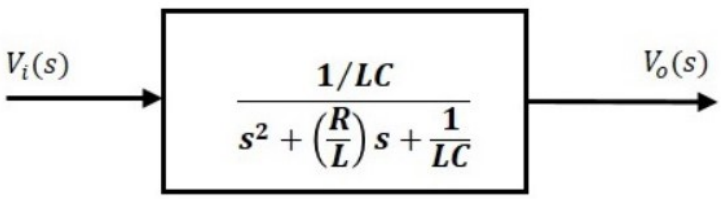
Transfer Function Model

$$i.e., Transfer\ Function = \frac{Y(s)}{X(s)}$$



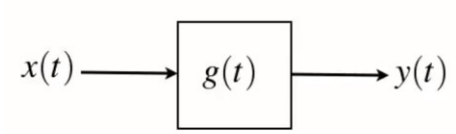
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Transfer Function Model

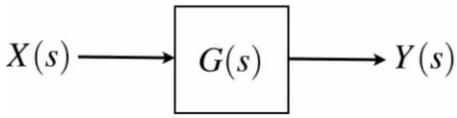


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Second Order Dynamic Systems



$$y(t) = x(t) \otimes g(t)$$



$$Y(s) = X(s)G(s)$$

$$G(s) = \frac{Y(s)}{X(s)}$$

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Example 2

To develop the mathematical model of the RC circuit, we need to derive the differential equation that describes its behavior. Let's consider a series RC circuit for this example.
The voltage across the resistor (V_R) and the voltage across the capacitor (V_{out}) add up to the input voltage (V_{in}).

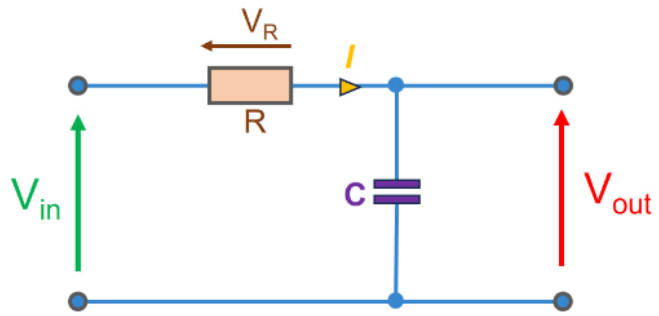


Figure 1: Schematic Diagram of the RC Circuit Configuration

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Example 2

Step 1: Applying Kirchhoff's voltage law (KVL) to the series RC circuit, we can write:

$V_{in} = V_R + V_{out}$

Step 2: Expressing the voltages using Ohm's law :

$V_{in} = I * R + V_{out}$

Step 3: Substituting I with $I = C * \frac{dV_{out}}{dt}$:

$V_{in} = (C * \frac{dV_{out}}{dt}) * R + V_{out}$

Step 4: Rearranging the equation and dividing both side with RC

$\frac{dV_{out}}{dt} + \frac{1}{RC} * V_{out} = \frac{1}{RC} * V_{in}$

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Example 2

Step 5: Defining the time constant $\tau = RC$

$\frac{dV_{out}}{dt} + \frac{1}{\tau} * V_{out} = \frac{1}{\tau} * V_{in}$

Step 6: Differential equation (Time domain)

$\tau * \frac{dV_{out}}{dt} = V_{in} - V_{out}$

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Step 7: Finally, converting the differential equation into a transfer function representation by taking the Laplace transform:

Where s is the Laplace variable and $V_{in}(0)$ is the initial voltage across the capacitor.

$$S * V_{out}(s) + \frac{1}{\tau} * V_{out}(s) = \frac{1}{\tau} * V_{in}(s)$$

Simplifying the equation further and solving for $\frac{V_{out}(s)}{V_{in}(s)}$, we get:

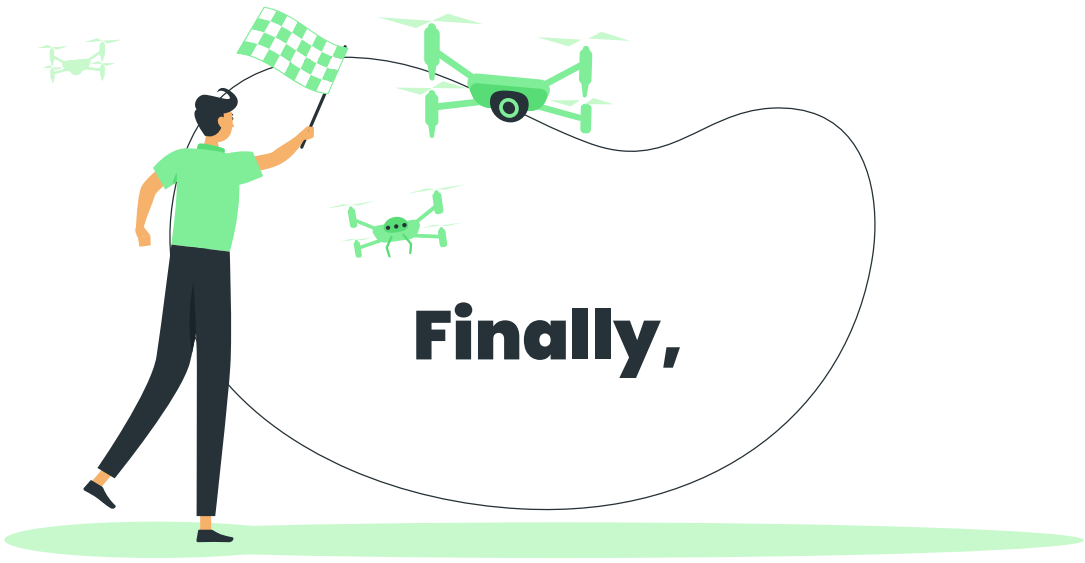
$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{1}{(S * \tau) + 1}$$

Step 8: Transfer Function (Frequency domain)

$$G(s) = \frac{1}{(S * \tau) + 1}$$

This is the transfer function representation of the RC circuit. It relates the input voltage $V_{in}(s)$ to the voltage across the capacitor $V_{out}(s)$ in the frequency domain.

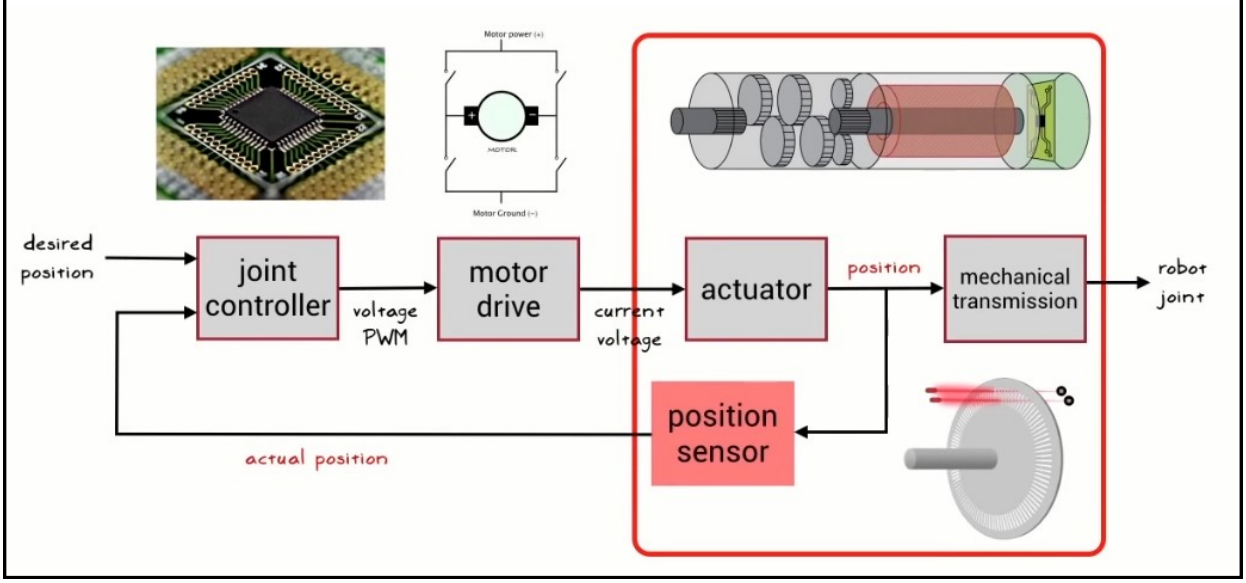
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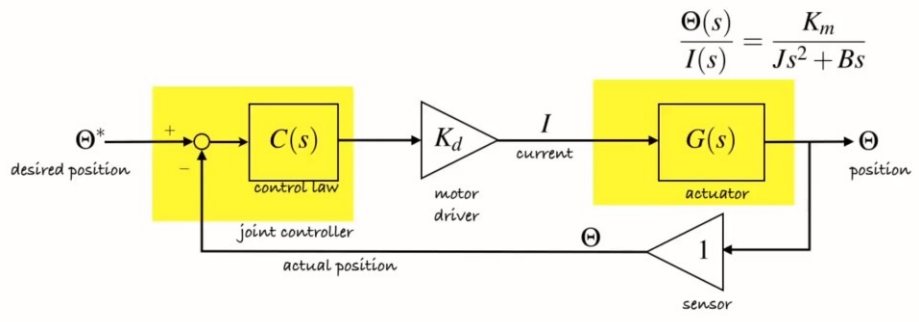
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Robot Joint Controller



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Joint Controller



- **The complete joint controller can be simulated using MATLAB Simulink**
- **Transfer function and PID controller blocks are readily available**

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Joint Controller

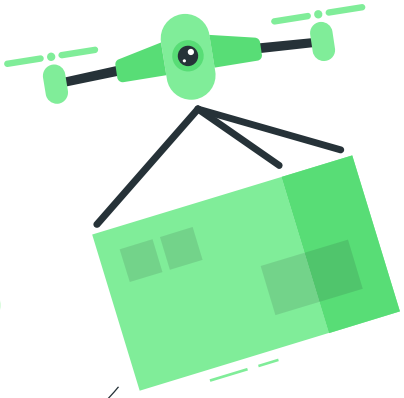
- Steps to design a controller
  - Mathematically model the system
  - Design Specifications
  - Use one of many control techniques: pole placement, root locus etc

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Robot Joint Controller

- Generally multiple joints
- **Common strategy**: design separate controller for each joint

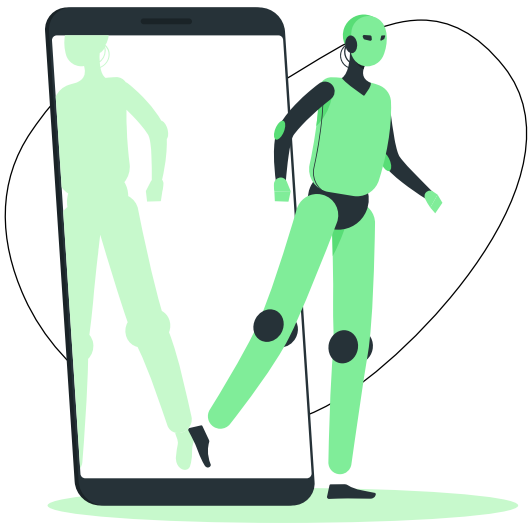
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Project Progress? Discussion

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Thanks!

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